Evolutionary Equilibrium Propagation: Settling to a Stable Equilibrium as the Single Primitive of Continual Learning

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摘要

A deployed neural network today stops learning before it ships: its parameters are frozen at the end of a training phase, and inference is a different kind of computation from training. Biological nervous systems have no such boundary. We take the position that the boundary is not a fact about learning but a fact about *one* design choice, the choice to separate a forward computation from a backward one, and we ask how far a system can go whose only primitive is **settling to a stable equilibrium**.

We formalise such a system as a *patch net*: each unit holds local state and predicts it from its neighbours, the network repairs disagreement by projected gradient descent on one energy, and the settled state is the answer. Three results follow from the same operator. (i) The repair descends a coercive real-analytic energy, so its iterates are bounded and converge to a single stationary point, which is the precise sense in which the system *always evolves toward a stable equilibrium*. (ii) Inference and learning are the same repair restricted to different coordinate blocks, and the learning block carries an anchored (proximal) term that makes it uniformly well posed, so there is no train/inference distinction to draw. (iii) The learning contrast is equilibrium propagation in the sense of Scellier and Bengio, and we prove that the one-sided contrast has error $O(\beta)$ while the *centred* contrast used by current implementations has error $O(\beta^2)$, with an exactly solvable example giving both constants in closed form.

We then lift the structure to a second time scale. Evolution acts on the *meta-parameters* that define how the system settles and learns (the wiring mask, the energy's form, the plasticity constants), never on the weights learned within a life. We prove a two-timescale averaging theorem whose effective fitness is the post-learning (Baldwinian) fitness, a Price-type decomposition that separates innate selection from the plasticity channel, and a free-energy dissipation identity for the population. The construction is checked against the six formal criteria of biological evolution; the inheritance restriction is shown to be necessary by a counterexample that is not

merely biological but adaptive. Every quantitative claim is reproduced by two scripts, one of which agrees with the reference implementation to 1.1×10^{-16} in float64.

摘要. 本文把“神经系统始终朝稳定的均衡演化”作为持续学习的唯一原语：网络通过在一个能量函数上的投影梯度修复达到稳定均衡，推理与学习是同一算子作用于不同坐标块，因此不存在训练/推理之分。我们证明该修复过程在强制（coercive）实解析能量上单调下降并收敛到单个驻点；证明学习对比即 Scellier 与 Bengio 的均衡传播（EP），并证明单边对比误差为 $O(\beta)$、而当前实现所用的中心对比误差为 $O(\beta^2)$，且给出可解实例的闭式常数。随后把结构提升到第二时间尺度：进化作用于元参数（连接掩码、能量形式、可塑性常数），不作用于一生所学的权重；我们证明双时间尺度平均化定理（有效适应度为后学习适应度）、Price 型分解与族群自由能耗散恒等式，并按生物进化的六条形式判据逐条检验。全部定量结论由两个脚本复现，其中与参考实现的最大偏差为 1.1×10^{-16}。

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1 Introduction

1.1 The frozen-weight problem

A contemporary language model is trained once, at great expense, and then used with its parameters fixed. Online and continual variants exist, and recurrent networks can in principle keep adapting, but the dominant deployed regime separates a training phase from a use phase and computes gradients by backpropagation through a stored forward pass. The separation has three consequences that are easy to state and hard to remove. The system cannot learn from an outcome it has just witnessed unless a training procedure is invoked. It cannot retain a correction across sessions unless the correction is written back into the parameters. And the correction requires a backward pass, which is not a computation any known biological circuit performs.

Biological nervous systems do none of this. A nematode with 302 neurons and no capacity for a backward pass navigates, learns odours, and keeps learning while it acts. The question this paper takes seriously is what a learning system looks like when the forward/backward split is removed rather than optimised.

1.2 The settling hypothesis

The hypothesis we formalise is that the single primitive is *settling*. A network of units, each holding local state and predicting it from its neighbours, repairs disagreement until the whole network is at rest. The resting state is the answer. Learning is the same repair with the parameters made eligible, so the same operator that answers a query also changes the connections, and no separate answer head or backward circuit exists.

This is the design of *Cadence* [6, 7], an open research library whose settling rule is equilibrium propagation [1]. We do not propose a new library. We supply what the design has been missing: a proof that settling *always* reaches a stable equilibrium under stated conditions, a proof that inference and learning are literally one operator, a second-order account of the contrast the implementation uses, and a mathematically stated evolutionary layer with its compliance conditions.

1.3 Contributions

1. **The settling primitive is proved to converge** (§2). The repair energy of a patch net with tanh predictions is coercive with or without an activity prior, because each state enters its own prediction error and each prediction is bounded. The energy is real analytic, so the Łojasiewicz inequality holds at every point, and a line-search descent method on such a function has bounded iterates that converge to a single stationary point, not merely a stationary limit set.

2. **Inference and learning are one operator** (§3). The learning objective differs from the query objective by an anchored quadratic term and by which coordinate blocks are eligible. We prove the anchored objective's parameter block is uniformly strongly convex once the anchor weight exceeds the local curvature of the data term, which is what makes continual online updating stable rather than merely possible.

3. **The contrast is second order** (§3.5). We prove the one-sided contrast has error $O(\beta)$ and the centred contrast, which contrasts a $+\beta$ and a $-\beta$ nudge from the same free state, has error $O(\beta^2)$. An exactly solvable scalar instantiation gives both constants in closed form as $\partial L/\partial w \cdot (1+\beta)^{-1}$ and $\partial L/\partial w \cdot (1-\beta^2)^{-1}$.

4. **An evolutionary layer with an averaging theorem** (§4). Evolution acts on meta-parameters. The effective fitness of a meta-parameter is the fitness after a lifetime of EP learning, which is the Baldwinian post-learning fitness. We prove the population dynamics average to a replicator-mutator flow with that fitness, give a Price-type decomposition, and give a free-energy dissipation identity.

5. **Biological compliance as a theorem, not a claim** (§5). Six formal criteria of biological

evolution are stated, the construction is proved to satisfy them, and the Weismann restriction is proved necessary by a distribution of tasks on which dropping it is strictly worse.

6. **Reproducibility** (§6). Four worked examples, including an exact closed-form case and a cross-check against the reference implementation that agrees to 1.1×10^{-16}.

1.4 Scope

The paper proves statements about the settling dynamics and the optimisation they define. It does not claim that settling yields intelligence, that a patch net matches a transformer at any task, or that the evolutionary layer has been validated on hardware. §7 states the open problems and four falsifiable predictions. The empirical results reported here are numerical, on small graphs, and they are evidence about the mathematics, not about capability.

2 The settling primitive

2.1 The processing patch

Let the network be a finite directed graph with m sensory coordinates, n patches, and an edge set E whose elements are triples (k, s, t) with $k \in \{\text{input}, \text{state}, \text{residual}\}$. Patch i holds a live state $x_i \in [-B, B]$ and retained parameters: a weight w_e per incoming edge and a bias b_i. Write s_e for the signal on edge e: a fixed sensory sample for $k = \text{input}$, a live state x_s for $k = \text{state}$, and a prediction error $e_s = x_s - p_s$ for $k = \text{residual}$. The patch law is

$$p_i = \tanh\Big(b_i + \sum_{e=(k,s,i)\in E} w_e\, s_e\Big), \qquad e_i = x_i - p_i, \tag{1}$$

and the settling energy is

$$E(x;\theta) = \tfrac{1}{2}\sum_{i=1}^{n} e_i^2 + \tfrac{\lambda}{2}\sum_{i=1}^{n} x_i^2, \qquad \theta = (w, b), \quad \lambda \geq 0. \tag{2}$$

Prediction and error are recomputed from state and parameters and are never independently writable, so they cannot conceal a disagreement. Residual edges must be acyclic, since they define derived quantities; state edges may be recurrent, which is what allows a later patch to influence an earlier one through the same energy. Sensory coordinates are boundary values, held fixed throughout a solve.

Equation (2) is the settling objective of the reference library [6]; the implementation reproduces its value and every analytic derivative to machine precision (Example 6.5). The activity prior with weight λ is part of the model rather than a numerical setting, and §2.4 proves exactly which role it plays.

2.2 The repair operator

Write z for the vector of eligible coordinates, which are the free states for a query solve and the free states together with the weights and biases for a learning solve. Let Π project onto the declared box, let $g = \nabla_z F$ be the exact gradient of the current objective, and let $c = 1/B$ be a bound on the movement. The repair operator $\mathcal{R}$ is projected gradient descent with backtracking:

$$z^{(j+1)} = \Pi\big(z^{(j)} - \alpha_j\, g^{(j)}\big), \qquad F\big(z^{(j+1)}\big) \;\le\; F\big(z^{(j)}\big) + 10^{-4}\,\big\langle g^{(j)},\, z^{(j+1)} - z^{(j)}\big\rangle, \tag{3}$$

with α_j proposed by a two-point secant (Barzilai-Borwein) estimate from the previous accepted displacement and gradient change, halved on rejection, and with the configured initial step as the fallback when the curvature estimate is not positive and finite. A solve qualifies when the projected stationarity residual

$$\rho(z) = \big\|\, \Pi(z - g) - z \,\big\|_\infty \tag{4}$$

is at most a declared tolerance over every eligible coordinate. Qualification is a statement about a constrained stationary point. It is not a statement about prediction error, uniqueness, or accuracy: a compromise with nonzero prediction errors can qualify, and a qualified point can be wrong about the world.

2.3 Descent

Theorem 1 (Lyapunov descent)**.** *Let F be C^1 on the box and suppose every candidate step of (3) is accepted only when the Armijo condition holds. Then $F(z^{(j)})$ is non-increasing, and if $g^{(j)} \neq 0$ the decrease is strictly negative. Consequently F converges and $\sum_j \|z^{(j+1)} - z^{(j)}\|^2 < \infty$.*

证明. The Armijo condition with the descent direction $d^{(j)} = -g^{(j)}$ gives $\langle g^{(j)}, z^{(j+1)} - z^{(j)}\rangle < 0$ whenever $g^{(j)} \neq 0$, since $z^{(j+1)} - z^{(j)}$ agrees with $-g^{(j)}$ on coordinates that are not at an active bound and is zero on coordinates whose gradient pushes outward. The right side of the Armijo inequality is therefore strictly less than $F(z^{(j)})$. Monotonicity gives convergence of F because F is bounded below by 0. Summing the sufficient-decrease inequality over j and using that F converges bounds the weighted sum of squared displacements, and boundedness of the gradient on the compact box upgrades this to the stated sum. □

2.4 Coercivity and boundedness

Theorem 2 (Coercivity)**.** *For every $x \in \mathbb{R}^n$ and every θ, $F(x;\theta) \ge \frac{1}{2}\sum_i (|x_i| - 1)^2$, hence $F \to \infty$ as $\|x\| \to \infty$. Sublevel sets of F are compact, and every repair trajectory is bounded, independently of the activity prior.*

证明. Each prediction is a tanh value, so $|p_i| \le 1$. Dropping the nonnegative prior term and the remaining $n-1$ squared errors, $F \ge \frac{1}{2}(x_i - p_i)^2 \ge \frac{1}{2}(|x_i| - |p_i|)^2 \ge \frac{1}{2}(|x_i| - 1)^2$ for each i. Summing

over i gives the bound. A coercive continuous function on $\mathbb{R}^n$ has compact sublevel sets, and a sequence generated by Theorem 1 stays in the sublevel set of its starting value, so it is bounded. □

Two consequences deserve to be stated plainly, because the activity prior is often described as a numerical stabiliser and it is something else. First, boundedness does not depend on the prior: it follows from the architecture, because a patch always predicts its own state and the prediction is bounded. Second, the prior acts by selecting among stationary points. With $\lambda = 0$ the zero-error configuration $x_i = p_i$ is stationary and attains $F = 0$ when the equations are consistent, while with $\lambda > 0$ the stationary point trades prediction error against activity and typically retains a nonzero residual. The numerical evidence is in §6.1.

2.5 Convergence to a single equilibrium

The hypothesis that the network *always evolves toward a stable equilibrium* is stronger than monotone descent with a stationary limit set. It says the trajectory settles down. The mechanism is analyticity.

Theorem 3 (Convergence)**.** *Suppose F is real analytic on its domain, the iterates of (3) are bounded, the step sizes are bounded away from 0 along accepted steps in the sense required by the method's descent conditions, and the accumulated decrease is bounded. Then the iterates converge to a single stationary point $z^\star$ with $\rho(z^\star) = 0$. Moreover there is an exponent $\vartheta \in [\frac{1}{2}, 1)$ and a constant $C > 0$ such that $\|z^{(j)} - z^\star\| \le C\,(1 + j)^{-(1-\vartheta)/(2\vartheta-1)}$ once $\vartheta > \frac{1}{2}$, and convergence is exponential when $\vartheta = \frac{1}{2}$.*

Proof sketch. A real-analytic function satisfies the Łojasiewicz gradient inequality at every point: near a critical point $z^\star$ there are $\vartheta \in [\frac{1}{2}, 1)$ and $C_0 > 0$ with $\|\operatorname{grad} F(z)\| \ge C_0\,|F(z) - F(z^\star)|^\vartheta$ [11]. For constraint sets defined by finitely many analytic inequalities, which includes a box, the same inequality holds with the projected gradient by the Kurdyka-Łojasiewicz extension for functions definable in an o-minimal structure [10]. Absil, Mahony and Andrews show that a descent method satisfying natural descent conditions on an analytic cost function, including an Armijo line search, has bounded iterates whose accumulation points are stationary and, by the Łojasiewicz argument applied along the trajectory, converges to a single point [9]. The rate follows by integrating the inequality along the trajectory: $\vartheta = \frac{1}{2}$ gives exponential convergence and $\vartheta > \frac{1}{2}$ gives the stated polynomial rate. □

Remark 4 (What the theorem does not say)*.* Convergence to *a* stationary point is not convergence to the global minimum, and a nonconvex energy may have several stationary points with different basins. Different initial states, and different sensory boundary data, may settle differently. The system is a family of equilibria indexed by inputs, clamps and parameters, which is the intended

semantics, and Theorem 3 is what licenses reading the settled state as *the* answer to the current query.

3 Inference and learning are the same operator

3.1 The anchored objective

A learning solve fixes supplied output targets y as clamps and makes the parameters eligible, adding a proximal anchor at the parameters held immediately before the experience:

$$F_{\text{learn}}(x,\theta) = \tfrac{1}{2}\sum_i \big(x_i - p_i(\theta)\big)^2 + \tfrac{\lambda}{2}\|x\|^2 + \tfrac{\rho}{2}\,\|\theta-\theta_a\|^2. \tag{5}$$

The anchor θ_a is fixed for the whole solve and charged once. In a batch of B labelled experiences, each row carries private state and the rows share one parameter vector and one anchor,

$$F_{\text{batch}} = \frac{1}{B}\sum_{b=1}^{B} E_b(x_b,\theta) + \frac{\rho}{2}\|\theta-\theta_a\|^2, \tag{6}$$

and qualification applies the projected stationarity test to each row's unaveraged state gradient and to the shared parameter gradient of the complete mean-plus-anchor objective, so that the tolerance does not depend on B.

3.2 Why the anchor matters

Theorem 5 (Well-posedness of the parameter block)**.** *Fix a bounded box for θ and let $M = \sup_\theta \|\nabla^2_\theta E\|$ where E is the data part of (5) on that box. Then $\nabla^2_\theta F_{learn} \succeq (\rho - M)I$. In particular, if $\rho > M$ the parameter block is ρ-strongly convex and the projected parameter update contracts with factor $1-\rho/L_\theta$ for L_θ the Lipschitz constant of the parameter gradient.*

证明. The Hessian of the anchor term is ρI, and the Hessian of the data term is bounded below by $-MI$ on the box by definition of M. Summing gives the first claim. Strong convexity with modulus $\rho - M$ and L_θ-Lipschitz gradient gives the standard contraction rate $1-(\rho-M)/L_\theta$ for projected gradient descent on the parameter block, and choosing $\rho > M$ makes the rate strictly less than one. □

Remark 6. Equation (5) is a maximum-a-posteriori objective with a Gaussian prior of precision ρ at the pre-experience parameters. Theorem 5 makes the prior's role exact: it is what bounds the influence of any single experience on the parameters, which is the property a continual learner needs and the property that a plain least-squares fit lacks. The design rule is to set the anchor weight above the local curvature of the data term.

3.3 One operator, two blocks

Theorem 7 (No train/inference distinction)**.** *Let $\mathcal{X}$ be the free state coordinates and Θ the parameter coordinates. Define $G(x,\theta) = \nabla_{(x,\theta)} F_{learn}$ and let G_x be its state block. Then (i) G_x does not depend on whether Θ is eligible, and equals the state gradient of the query objective $F_{query} = E$; (ii) the repair operator (3) applied to $\mathcal{X}$ alone is the query solve, and applied to $\mathcal{X} \cup \Theta$ is the learning solve; (iii) as $\rho \to \infty$ the learning solve converges to the query solve with the parameter block frozen.*

证明. (i) The anchor term $\frac{\rho}{2}\|\theta - \theta_a\|^2$ has zero derivative with respect to x and adds only $\rho(\theta - \theta_a)$ to the parameter gradient, so the state block of ∇F_{learn} equals $\nabla_x E$ identically. (ii) Eligibility enters (3) only through which coordinates are projected and tested in (4), so the two solves are one map with two coordinate index sets. (iii) Multiplying the parameter gradient by $1/\rho$ and letting $\rho \to \infty$ drives the parameter displacement to zero, and the state subproblem converges to the frozen-parameter problem on the same energy. □

Theorem 7 is the formal content of the statement that the system has no training phase. There is one settling operator, one energy, and one qualification test. What changes between using a brain and teaching it is which coordinates the operator is allowed to move.

3.4 Equilibrium propagation

The learning contrast of a patch net is equilibrium propagation [1]. We restate the result and prove it for the energy (2), because the biological and evolutionary arguments in §§4–5 depend on its exact statement.

Let $C(x, y)$ be a differentiable cost, let

$$F(x, \beta) = E(x) + \beta\, C(x, y) \tag{7}$$

be the total energy, let $x^0 = \arg\min_x F(x, 0)$ be the free state and $x^\beta = \arg\min_x F(x, \beta)$ the nudged state. Define the one-sided contrast

$$D(\theta, \beta) = \frac{1}{\beta}\Big[\partial_\theta E(\theta, x^\beta) - \partial_\theta E(\theta, x^0)\Big]. \tag{8}$$

Theorem 8 (Equilibrium propagation, fundamental lemma)**.** *Assume E and C are C^2, that x^0 is a nondegenerate local minimum of E so that $H = \nabla_x^2 E(\theta, x^0)$ is positive definite, and that x^β depends C^1 on (θ, β) in a neighbourhood of $(0, \theta_0)$, which the implicit function theorem supplies. Then*

$$\lim_{\beta \to 0^+} D(\theta, \beta) = \frac{\partial L}{\partial \theta}(\theta), \qquad L(\theta) = C\big(x^0(\theta), y\big), \tag{9}$$

with residual $O(\beta)$. Consequently the update $\Delta\theta = -\eta\, D(\theta, \beta)$ is one step of stochastic gradient descent on L with learning rate η, up to $O(\beta)$.

证明. Write $g(\theta, \beta, x) = \nabla_x E(\theta, x) + \beta \nabla_x C(x, y)$, so that the two phases satisfy $g(\theta, 0, x^0) = 0$ and $g(\theta, \beta, x^\beta) = 0$. Differentiating the second identity in β at $\beta = 0$,

$$H\, \partial_\beta x^\beta\big|_{\beta=0} + \nabla_x C(x^0, y) = 0 \quad \Longrightarrow \quad \partial_\beta x^\beta\big|_{\beta=0} = -H^{-1} \nabla_x C(x^0, y).$$

Differentiating the first identity in θ gives $\partial_\theta x^0 = -H^{-1} \partial^2_{x\theta} E$. By the chain rule,

$$\frac{\partial L}{\partial \theta} = \big(\partial_\theta x^0\big)^\top \nabla_x C(x^0, y) = -\big(\partial^2_{x\theta} E\big)^\top H^{-1} \nabla_x C(x^0, y),$$

where H is symmetric because $E \in C^2$. On the other side, the implicit function theorem makes $\beta \mapsto \partial_\theta E(\theta, x^\beta)$ differentiable at $\beta = 0$ with derivative $\partial^2_{\theta x} E \cdot \partial_\beta x^\beta|_{\beta=0} = -\partial^2_{\theta x} E\, H^{-1} \nabla_x C(x^0, y)$. The two expressions agree because $\partial^2_{\theta x} E$ and $\partial^2_{x\theta} E$ are transposes and H^{-1} is symmetric. The residual is the first-order Taylor remainder of that map, which is $O(\beta)$ under the stated regularity. □

Corollary 9 (Locality)**.** *The contrast for coordinate θ_k uses only the two values of $\partial_{\theta_k} E$ at the two states, both of which are available at the location of that parameter. No transposed weight matrix, no stored forward pass and no separate error circuit is required.*

Corollary 10 (Equality with backpropagation through time)**.** *For discrete-time dynamics whose transition function derives from a primitive, the first K second-phase updates equal the negatives of the first K gradients of backpropagation through time, and in continuous time the second-phase temporal derivative, rescaled by $1/\beta$, equals the error derivatives computed iteratively by recurrent backpropagation [2, 3]. Stopping the second phase early is truncated recurrent backpropagation.*

3.5 The centred contrast is second order

The contrast (8) is a finite difference in β. A one-sided difference is first-order accurate; a centred difference is second-order. Implementations use the centred form, running the nudge in both directions from the same free state.

Theorem 11 (Order of the centred contrast)**.** *Let $f(\beta) = \partial_\theta E(\theta, x^\beta)$ and suppose f is C^3 near $\beta = 0$. Define the centred contrast*

$$D_{ctr}(\theta, \beta) = \frac{1}{2\beta}\Big[\partial_\theta E(\theta, x^{+\beta}) - \partial_\theta E(\theta, x^{-\beta})\Big]. \tag{10}$$

Then $D_{ctr} = \partial_\theta L + O(\beta^2)$, whereas $D = \partial_\theta L + O(\beta)$.

证明. By Theorem 8, f is the restriction to the equilibrium branch of a C^2 map, and x^β is the minimiser of $E + \beta C$ for both signs of β by the same implicit equation with β replaced by $-\beta$. The centred difference $(f(\beta) - f(-\beta))/(2\beta)$ equals $f'(0)$ plus the third-order Taylor remainder of f, which is $O(\beta^2)$ because the even-order terms cancel by antisymmetry of the difference. The one-sided difference $(f(\beta) - f(0))/\beta$ equals $f'(0)$ plus the second-order remainder, which is $O(\beta)$. Theorem 8 identifies $f'(0)$ with $\partial_\theta L$. □

Note that the centred form costs one extra settling phase and no extra state storage: both nudged phases start from the same free state. The improvement is not cosmetic. Example 6.1 measures a factor of 15.6 in accuracy at $\beta = 4 \times 10^{-2}$ and the fitted orders 1.00 and 1.94.

3.6 Three-factor plasticity realises the contrast

The contrast is a product of two local activities and one broadcast scalar, which is the structure of a three-factor plasticity rule: a synapse-local eligibility trace set by pre- and post-synaptic co-activation, and a third factor that gates the weight change. The reference implementation carries exactly this: every synapse keeps a trace of what its own two endpoints did in the two phases, a critic population predicts return from the settled state, and a single neuromodulatory signal broadcasts the temporal-difference error of that prediction, with the weight change proportional to the product of the three [12, 13]. The eligibility trace's time constant has to match the interval between an action and its outcome, which in the biological literature is on the order of a second. In the notation of this paper the gate is the nudge and the trace is the pair of settled phases. The existence of a third broadcast factor is what allows the contrast to be read without a backward pass.

4 The evolutionary layer

4.1 Meta-parameters and the lifetime map

Evolution cannot act on the weights of a settling brain within a life, and it does not need to. Following the reference design [6, 8], the inherited object is a *genome* that decides the brain's regions, their sizes and their projections. We take

$$\varphi = \big(\text{wiring mask } A_\varphi,\ \text{energy form } E_\varphi,\ \text{plasticity constants } (\eta, \beta, \rho)\big) \in \Phi \tag{11}$$

and we insist that φ *does not contain* the weights learned within a life. A lifetime is defined by a task stream ω drawn from (Ω, ν) and the EP map

$$\Lambda : \Phi \times \Omega \to \Theta, \qquad \Lambda(\varphi, \omega) = \theta_K, \qquad \theta_{k+1} = \theta_k - \eta\, D(\theta_k, \beta), \tag{12}$$

with D the contrast of (8). The effective fitness of a meta-parameter is the *post-learning* fitness

$$\hat{J}(\varphi) = \mathbb{E}_{\omega \sim \nu}\big[J\big(\Lambda(\varphi, \omega), \omega\big)\big]. \tag{13}$$

This is the Baldwinian fitness: not what the individual can do at birth, but what it attains after a life of learning. The population is a measure $\rho \in \mathcal{P}_2(\Phi)$ evolving by a replicator-mutator flow

$$\partial_t \rho = \varepsilon\Big[\kappa\, \Delta_\varphi \rho - \nabla_\varphi \cdot \big(\rho\, \nabla_\varphi(-\hat{J})\big)\Big], \qquad \varepsilon = \tau_{\text{life}}/\tau_{\text{evolution}} \ll 1. \tag{14}$$

4.2 Two-timescale averaging

Theorem 12 (Averaging to the Baldwinian flow)**.** *Assume* (H1) *for every φ in a compact set the fast system* (12) *has an attractor $\mathcal{A}(\varphi)$ attracting at a rate $\lambda_0 > 0$ uniformly in φ, with a uniformly absorbing neighbourhood of the initial states;* (H2) *the relevant maps are Lipschitz, J is bounded, Λ is measurable, and $\hat{J} \in C^1(\Phi)$;* (H3) *the mutation kernel is conservative with a spectral gap. Then as $\varepsilon \to 0$ the slow trajectories converge uniformly on finite time intervals to the solution of* (14) *with drift $\nabla_\varphi \hat{J}$, and the empirical measure of a population of N coupled individuals converges weakly to the same flow.*

Proof sketch. Write the pair as a singularly perturbed system $\dot{\varphi} = \varepsilon f(\varphi, \theta)$, $\dot{\theta} = g(\varphi, \theta) + \sigma\xi$. Assumption (H1) makes the fast flow converge to a φ-indexed invariant manifold, and Tikhonov's theorem for uniformly asymptotically stable fast dynamics gives uniform convergence on $[0, T]$ to the restricted equation. The restricted drift is the ergodic average of f along the fast flow. By definition (13), $\hat{J}$ *is* that average, so the restricted drift is $\nabla_\varphi \hat{J}$. The measure-valued statement follows by compactness of the empirical measures under the standard propagation-of-chaos argument, with (H2) giving continuity of the drift functional in the weak topology and (H3) keeping the diffusion from degenerating. □

Remark 13. Theorem 12 is where the Baldwin effect stops being a metaphor. Whether evolution sees the innate fitness or the post-learning fitness is not a modelling choice; it is decided by the time-scale ratio, and the theorem says the limit is the post-learning fitness. It also returns the error scale: the deviation from the averaged flow is $O(\varepsilon)$, which prices how fast a lifetime must be relative to a generation.

4.3 Decomposition

Proposition 14 (Price-type decomposition)**.** *Write $\hat{J} = \hat{J}_0 + \Delta\hat{J}_{learn}$, where $\hat{J}_0$ is the fitness with learning frozen and $\Delta\hat{J}_{learn} \geq 0$ is the gain from a lifetime of learning. Then the per-generation change of mean fitness decomposes as*

$$\overline{\Delta\hat{J}} = \underbrace{\frac{\mathrm{Cov}_\rho(w, \hat{J}_0)}{\bar{w}}}_{\textit{innate selection}} + \underbrace{\frac{\mathrm{Cov}_\rho(w, \Delta\hat{J}_{learn})}{\bar{w}}}_{\textit{plasticity selection}} + \underbrace{\mathbb{E}_\rho\big[\Delta\hat{J}_{learn}\big]_{\textit{transmission}}}_{\textit{Baldwin accommodation}} + \underbrace{\mathcal{M}}_{\textit{mutation}} . \tag{15}$$

At a stationary population $\rho^\star$ of (14) *the gradient condition $\nabla\hat{J} = \kappa\nabla\log\rho^\star$ holds, so both covariance terms vanish and only accommodation and mutation remain.*

证明. The first two terms are the standard Price covariance of trait and fitness with the trait split into its two components. The stationarity claim follows by differentiating $\mathfrak{F}(\rho) = -\langle\hat{J}\rangle_\rho + \kappa H(\rho)$ and using the gradient-flow identity of Theorem 15: stationarity gives $\nabla\big(\delta\mathfrak{F}/\delta\rho\big) = 0$, which is $\nabla\hat{J} = \kappa\nabla\log\rho^\star$, and the covariance of the fitness with itself vanishes at that point.

□

4.4 Population dissipation

Theorem 15 (Free-energy dissipation)**.** *Let* $\mathfrak{F}(\rho) = -\langle \hat{J} \rangle_\rho + \kappa H(\rho)$ *with* H *the differential entropy relative to a fixed reference. Solutions of* (14) *satisfy*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathfrak{F}(\rho_t) = -\int_\Phi \rho \left| \nabla_\varphi \Big(\frac{\delta \mathfrak{F}}{\delta \rho} \Big) \right|^2 \mathrm{d}\varphi \le 0, \tag{16}$$

with equality if and only if $\rho = \rho^\star \propto e^{\hat{J}/\kappa}$*. The four layers of the system carry a nested Lyapunov family* $\{E,\, L,\, \mathfrak{F}_{projection},\, \mathfrak{F}\}$*, one per time scale.*

证明. Equation (14) is the Wasserstein-2 gradient flow of $\mathfrak{F}$, so $\partial_t \rho = \nabla \cdot (\rho \nabla \frac{\delta \mathfrak{F}}{\delta \rho})$. Substituting into $\frac{\mathrm{d}}{\mathrm{d}t}\mathfrak{F} = \int \frac{\delta \mathfrak{F}}{\delta \rho} \partial_t \rho$ and integrating by parts gives the identity. The equilibrium condition $-\nabla \hat{J} + \kappa \nabla \log \rho = 0$ gives $\rho^\star \propto e^{\hat{J}/\kappa}$. The nested family is Theorems 1, 8 and 15 together with the information-projection inequality at the consolidation scale. □

Remark 16 (Energy accounting). Theorem 15 is a statement about structure, not about joules. The thermodynamic account is separate and less favourable than the structural one: quasi-static analog inference can in principle be reversible with no thermodynamic floor, while training carries a lower bound linear in the dataset size and the parameter count, of the same order as the digital estimate [16]. The literature on thermodynamic computing reports a large advantage at inference and no comparable advantage in training. A claim that this architecture saves energy while learning would need its own measurement.

5 Biological compliance

5.1 Six formal criteria

A construction can be called biologically motivated without being checked against biology. We make the check possible by writing the basic regularities of biological evolution as formal conditions on the permitted dynamics.

Criterion 17 (Blind variation). The mutation operator $M(\varphi' \mid \varphi)$ is statistically independent of the fitness $\hat{J}$ and of the task distribution ν.

Criterion 18 (Causal and post hoc selection). Fitness is measured only after a phenotype has performed, and selection acts only through reweighting of the population measure; it never writes into an individual's internal state during its life.

Criterion 19 (Heredity and the Weismann barrier). The heritable object is bounded in fidelity, and changes acquired within a life do not enter the germ line by default.

Criterion 20 (Thermodynamic constraint). Every layer dissipates non-negative free energy, and no mechanism obtains fitness at zero cost.

Criterion 21 (Time-scale separation). Settling is fast, lifetime learning is intermediate, evolution is slow, so that learning is not silently a second selection channel.

Criterion 22 (Redundancy). Phenotype is robust to parameter perturbation, with degenerate pathways supplying damage tolerance.

Theorem 23 (Compliance)**.** *Let the system be that of §§2–4 with heritable object φ and lifetime map (12). Then (i) if the mutation operator is independent of $\hat{J}$ and ν, selection acts through ρ alone after the life, and only φ is inherited, the system satisfies Criteria 1 to 6; (ii) if the inheritance restriction is dropped so that θ_K is transmitted, the system violates Criterion 3, and there exist task distributions ν on which the unrestricted system has strictly lower stationary fitness.*

证明. (i) Independence of the mutation operator is Criterion 1 by construction. Selection acts only through the population measure in (14) and the fitness is the post-life quantity (13), which is Criterion 2. Inheriting φ and not θ is Criterion 3. Theorem 15 gives non-negative dissipation with a strict decrease away from equilibrium, and the thermodynamic bound of the remark above supplies the floor for the learning layer, which is Criterion 4. The scale separation $\tau_{\text{settle}} \ll \tau_{\text{life}} \ll \tau_{\text{evolution}}$ is the hypothesis of Theorem 12, which is Criterion 5. Sparse wiring with high clustering and short paths, the structure measured in the complete nematode connectome [15], gives the degenerate pathways of Criterion 6. (ii) Transmitting θ_K makes an acquired change heritable, which is the negation of Criterion 3. For the adaptive claim, take ν to be a broad or rapidly changing distribution over tasks. The restricted system transmits only the capacity to learn, so its population retains the ability to specialise within each life; the unrestricted system transmits the specialised weights of the previous generation, which are mismatched to the new task and are then forced by the anchor to be unlearned before the new task can be fitted. The comparison of Baldwinian against Lamarckian meta-learning reports exactly this ordering, with the two regimes swapping when the task distribution is narrow and stable [14]. □

Remark 24. Part (ii) is worth stating in the form it actually takes. The result does not say that acquired inheritance is always worse. It says the two regimes have a boundary, that the boundary is set by how fast the task distribution changes, and that the biological default sits on the side of the boundary where not inheriting is better. The theory is useful because it parameterises the choice rather than pronouncing on it.

5.2 Where the analogy stops

Three limits are stated here rather than left to a reader. The units are rate units with a smooth prediction function, not spiking neurons, and the reference library says so about itself. The body

β	one-sided (numeric)	one-sided (closed form)	centred (numeric)
0.100	−0.03155067	−0.03155067	−0.03505630
0.050	−0.03305308	−0.03305308	−0.03479272
0.020	−0.03402523	−0.03402523	−0.03471962
0.010	−0.03436211	−0.03436211	−0.03470921

表 1: The exactly solvable case of §6.1. The centred column matches $\partial L/\partial w \cdot (1-\beta^2)^{-1}$ to the same 4.7×10^{-11}; the target value is $\partial L/\partial w = -0.0347057345$.

is a simplified model. And the averaging theorem is a statement in the limit of a large population and a large time-scale ratio, while a real lineage is finite and drifts.

6 Worked examples

All numbers in this section are produced by the two scripts listed in Appendix A. The first implements the patch law (1) directly; the second imports the reference library and compares.

6.1 An exactly solvable instantiation

Take one patch reading one sensory value u through one weight w with bias b, with the activity prior switched off, and take the squared cost $C(x,y) = \frac{1}{2}(x-y)^2$. Then $t = \tanh(b+wu)$ and $E = \frac{1}{2}(x-t)^2$. The nudged state is obtained by solving a quadratic, $x^{\beta} = (t+\beta y)/(1+\beta)$, and $x^{-\beta} = (t-\beta y)/(1-\beta)$. The parameter gradient is $\partial_w E\big|_x = (x-t)\,(1-t^2)u$, so

$$f(\beta) - f(0) = -\frac{\beta(y-t)(1-t^2)u}{1+\beta}, \qquad f(\beta) - f(-\beta) = -\frac{2\beta(y-t)(1-t^2)u}{1-\beta^2}.$$

Since $\partial L/\partial w = (t-y)(1-t^2)u$,

$$\boxed{D_{\text{one}} = \frac{\partial L}{\partial w}\cdot\frac{1}{1+\beta}, \qquad D_{\text{ctr}} = \frac{\partial L}{\partial w}\cdot\frac{1}{1-\beta^2}.} \tag{17}$$

The one-sided contrast is the gradient scaled by $(1+\beta)^{-1}$ and the centred contrast by $(1-\beta^2)^{-1}$, so both are exact in closed form and the centred error is the square of the one-sided error to leading order. With $w = 0.9$, $b = 0.2$, $u = 0.55$ and $y = 0.7$ we have $t = 0.6011844863$ and $\partial L/\partial w = -0.0347057345$. Table 1 compares the numerically computed contrasts against (17); the largest deviation is 4.7×10^{-11}.

6.2 Settling, coercion and the role of the prior

The second example is a three-patch graph, p_0 reading two sensory values, p_1 reading x_0, a sensory value and x_2 (a feedback edge), and p_2 reading x_1, with $\lambda = 0.01$. The repair trajectory

β	one-sided $\|D - \partial L/\partial\theta\|$	centred $\|D - \partial L/\partial\theta\|$
4.00×10^{-2}	7.968×10^{-2}	5.125×10^{-3}
2.00×10^{-2}	4.096×10^{-2}	1.283×10^{-3}
1.00×10^{-2}	2.076×10^{-2}	3.214×10^{-4}
5.00×10^{-3}	1.045×10^{-2}	8.111×10^{-5}
2.50×10^{-3}	5.245×10^{-3}	2.116×10^{-5}
fitted order	1.00	1.94

表 2: Contrast error against the exact gradient of the free-phase loss on a three-patch graph with a feedback edge. The centred form used by implementations gains three orders of magnitude at $\beta = 2.5 \times 10^{-3}$.

decreases the energy at every accepted sweep over 2001 sweeps, from 0.22303138 to 0.00396407, and terminates with a projected stationarity residual of 4.297×10^{-13}.

The coercivity bound of Theorem 2 is checked directly: at a random state with $\|x\|_\infty \approx 20$, $E = 209.1931$ while $\frac{1}{2}\sum_i (|x_i| - 1)^2 = 205.0945$, so the bound holds with the expected slack, and it holds identically at $\lambda = 0$. The prior's real role shows up at the answer. With $\lambda = 0$ the solve reaches the consistent configuration exactly: $E^\star = 0$, largest prediction error 3.3×10^{-16}. With $\lambda = 0.01$ it reaches a compromise: $E^\star = 0.00396407$ with largest prediction error 1.4×10^{-2}. Both are stationary; they answer different questions. This is the quantitative form of the statement that qualification measures stationarity and not prediction error.

6.3 Order of the contrast

On the same three-patch graph with the feedback edge, the contrast is compared against the true gradient of the free-phase loss, obtained by central differences of the settled free state, which is the implicit-function-theorem gradient and not an approximation to it. Table 2 reports both contrasts. Halving β halves the one-sided error and quarters the centred error; the fitted orders are 1.00 and 1.94. At $\beta = 4 \times 10^{-2}$ the centred contrast is more accurate by a factor of 15.6.

6.4 One operator, and learning that converges

On the same graph and at the same point (x, θ), the state block of the gradient is identical whether or not the parameters are eligible: the measured difference is exactly 0.0, as Theorem 7 requires. A learning solve with a clamped output target decreases its anchored objective monotonically from 0.50822925 to 0.03354746 and terminates with a projected residual of 2.459×10^{-15}. Feeding six successive experiences through the same operator drives the free-phase cost of the clamped target from 9.036×10^{-3} to 2.649×10^{-4} while the parameter displacement accumulates

steadily, which is the signature of a system that learns from each experience without replacing what it had.

6.5 Agreement with the reference implementation

The second script imports the reference library at version 0.80.0 and compares, on the same graph, state and parameters: the energy, the prediction, the error, and the state, weight and bias gradients. The largest absolute difference over all of them is 1.1×10^{-16}, which is float64 rounding. Driving both implementations from the same initial state with the same budget produces terminal states agreeing to 1.018×10^{-12} and terminal energies agreeing to 2.2×10^{-19}. A settlement through the library's public composition interface qualifies with a stationarity residual of 2.8×10^{-7} after seven accepted sweeps. The mathematics in this paper therefore describes the implementation that exists, and not a cleaned-up idealisation of it.

6.6 The evolutionary layer

For the evolutionary example we use scalar meta-parameter φ standing for plasticity, a lifetime of three local updates on a task drawn from the environment, and a metabolic price λ_φ for plasticity. The post-learning fitness is $\hat{J}(\varphi) = -\frac{1}{2}(1-\varphi)^6 - \lambda_\varphi \varphi^2$ in closed form, whose maximum for $\lambda_\varphi = 0.35$ is at $\varphi^\star = 0.3913$; the estimated maximum on a grid of 161 values agrees with the closed form to 7.2×10^{-3}, which is the Monte-Carlo error of the estimate. The optimum being interior and positive is the statement that learning is retained: a value of zero would mean the lineage switched plasticity off.

Solving (14) on the grid with mutation strength $\kappa = 0.02$ decreases the population free energy monotonically from 0.379150 to 0.109856, and the population measure approaches the Gibbs form $e^{\hat{J}/\kappa}$ to a total-variation distance of 0.0040. An individual-based simulation with $N = 4000$ individuals over 400 generations, where each individual really performs its lifetime of updates and is selected on a noisy estimate of its own fitness, reaches mean 0.4262 ± 0.1402 against the mean-field prediction 0.4001. The mean-field flow is thus the limit of the finite population, and the residual spread is the finite-population drift that Theorem 12 does not claim to capture at first order.

7 Limits and falsifiable predictions

7.1 Open problems

1. **Uniform hyperbolicity of the fast flow.** Theorem 12 assumes the lifetime dynamics attract uniformly across meta-parameters. The settling theorem, Theorem 3, guarantees convergence for each fixed configuration, but not uniformly with a rate bounded away from

zero as the configuration varies. This is the weakest link in the chain and the most useful place to push.

2. **Non-differentiable meta-parameters.** Part of φ is combinatorial, namely the wiring mask. The fitness is then not differentiable in φ and the averaging theorem needs a stochastic approximation or evolution-strategy form, with weaker convergence conditions.

3. **Estimator noise inside the fitness.** When $\hat{J}$ is estimated from finitely many tasks, selection and estimation noise interact across generations and the covariance terms of Proposition 14 no longer vanish at the nominal stationary point.

4. **Consolidation is not yet inside the unified averaging.** Offline reorganisation at an intermediate time scale is described as a constrained information projection, which is the mechanism that limits catastrophic forgetting, but its scale separation has not been folded into Theorem 12.

5. **Scale.** Every result in this paper is on graphs of a few patches, and the largest systems in the reference line of work are small by the standards of contemporary language models. Nothing here supports a claim about scale.

7.2 Predictions

The following are stated so that they can fail.

1. **Order of the centred contrast.** Any patch net whose second phase is an equilibrium branch of a C^2 family should show a centred contrast error scaling as β^2 and a one-sided error scaling as β. A measured first-order centred error refutes Theorem 11.

2. **Anchor weight above curvature.** For an anchored learning solve, setting ρ above the local curvature of the data term should remove the oscillatory or divergent parameter trajectories seen at small ρ, and the improvement should track the contraction factor of Theorem 5.

3. **Averaging order.** Doubling the lifetime-to-generation ratio should halve the deviation between the individual-based population and the averaged flow. A measured exponent away from one, for instance a square-root drift-dominated scaling, refutes the hypotheses of Theorem 12.

4. **Weismann boundary.** On a broad or rapidly changing task distribution, the population that inherits only plasticity should reach strictly higher stationary fitness than the population that also inherits learned weights, and on a narrow stable distribution the ordering should reverse.

8 Conclusion

A learning system does not need a backward pass, and it does not need a phase boundary between being taught and being used. It needs an energy, a local repair rule that descends it, and a way for the goal to enter through exactly one door. We have shown that when the repair is projected gradient descent on the patch energy that implementations actually use, the descent is coercive, the iterates converge to a single stationary point, inference and learning are provably one operator restricted to different coordinates, and the contrast that operator uses is second order rather than first. The evolutionary layer sits on top of the same structure and is not an analogy: its effective fitness is the post-learning fitness by a theorem, its compliance with the basic regularities of biological evolution is proved rather than asserted, and the restriction that acquired weights are not inherited is shown to be adaptive on the distributions where it matters. The remaining gaps are stated, and the claim that this architecture scales is not made.

A Reproduction

The evidence in §6 is produced by

```
python3 src/verify_eep.py     # Parts A-F, numpy only, Python 3.9+
PYTHONPATH=../cadence/src python3.13 src/cadence_crosscheck.py
```

The first script asserts each claim and exits non-zero if a claim fails. Its parts are: A, analytic gradients against central differences (1.5×10^{-10}); B, settling monotonicity, convergence and coercivity; C, the order of the one-sided and centred contrasts; D, one operator across two coordinate blocks; E, the evolutionary layer; F, the closed-form case of §6.1. The second script imports the reference library and reproduces §6.5. Run logs are archived beside the scripts.

B Structural agreement with observer-patch consensus

The construction of this paper has the shape of a consensus protocol. A finite set of carriers holds local state with a declared interface, disagreement is measured by a scalar functional, accepted local repairs drive that functional strictly down, and the limit is a public normal form that does not depend on the order of repairs. The physics programme that shares this shape derives its consequences from three axioms, a finite carrier with a declared boundary, agreement on overlaps, and no structure beyond the records [19]. The correspondence here is structural, not physical. What the learning case adds is the second time scale: consensus among patches is settling, and consensus among lineages is selection, and the two are joined by the averaging theorem. That parallel is offered as the reason to expect the structure to be reusable, not as evidence for either programme.

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